A finite-history obstruction to Lyapunov certificates for alternating gradient descent–ascent

Abstract

A recent study of projected alternating gradient descent–ascent proposes a homogeneous finite-history Lyapunov linear matrix inequality and reports approximate numerical feasibility. We prove that its exact feasible set is empty for every finite history and every positive stepsize. More directly, for each history length we construct a genuine initialized trajectory on compact convex polytopes, with a symmetric positive definite norm-one operator, whose two consecutive history Gram matrices agree while an admissible comparator gap is positive. Hence the specified quadratic history potential cannot satisfy the required one-window descent inequality, even without a positivity requirement on the potential. The construction uses an exact finite sine-moment system and normal-cone inequalities. This obstruction does not disprove an averaged convergence rate for the algorithm, the source's conditional theorem, or its separate matrix-game theorem.

Keywords: alternating gradient descent–ascent; Lyapunov certificate; semidefinite programming; convex projection

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1 The certificate question

For nonempty compact convex sets $X \subset \mathbb{R}^n$ and $Y \subset \mathbb{R}^m$, a linear map $A : \mathbb{R}^n \to \mathbb{R}^m$, and a fixed $\eta > 0$, projected alternating gradient descent–ascent generates

$$x_{k+1} = P_X(x_k - \eta A^T y_k), \qquad y_{k+1} = P_Y(y_k + \eta A x_{k+1}), \qquad k \geq 0. \tag{1}$$

Here P_C is Euclidean projection and $(x_0, y_0) \in X \times Y$ is arbitrary. With an arbitrary feasible comparator $(x_\star, y_\star)$, define

$$p_k = Ax_k, \quad q_k = A^T y_k, \quad p_\star = Ax_\star, \quad q_\star = A^T y_\star, \qquad g_k = \langle q_\star, x_k \rangle - \langle p_\star, y_k \rangle.$$

The comparator is not assumed to be a saddle point. This is the quantifier required to recover the duality gap by taking a supremum after telescoping.

Write the actual projection residuals as

$$\bar{d}_{k+1} = x_k - \eta q_k - x_{k+1} \in N_X(x_{k+1}), \qquad d_{k+1} = y_k + \eta p_{k+1} - y_{k+1} \in N_Y(y_{k+1}), \tag{2}$$

where $N_C(z) = \{v : \langle v, u - z \rangle \leq 0 \text{ for all } u \in C\}$, and put $\bar{d}_0 = d_0 = 0$. This last convention pads only the initialized state; a later base window contains the actual residuals.

For an integer $h \geq 0$, consider exactly the two history matrices used in Appendix B of Nan et al. [1]:

$$\begin{aligned} H_k^x &= [x_\star \mid q_\star \mid x_k \mid \bar{d}_k \mid \cdots \mid \bar{d}_{k+h+1} \mid q_k \mid \cdots \mid q_{k+h}], \\ H_k^y &= [y_\star \mid p_\star \mid y_k \mid d_k \mid \cdots \mid d_{k+h+1} \mid p_k \mid \cdots \mid p_{k+h+1}]. \end{aligned} \tag{3}$$

They have $2h+6$ and $2h+7$ columns. For symmetric matrices Q_x, Q_y of these respective sizes, the proposed potential is

$$V_k = \operatorname{tr}\big(Q_x (H_k^x)^T H_k^x\big) + \operatorname{tr}\big(Q_y (H_k^y)^T H_k^y\big). \tag{4}$$

The desired descent condition is

$$V_{k+1} - V_k + g_k \leq 0 \tag{5}$$

on every actual window, with the same coefficients for the normalized class $\|A\| \leq 1$.

Theorem 2 of the 26 September 2026 preprint [1] derives an $O(1/T)$ averaged-gap bound *conditional* on exact feasibility of a specific Lyapunov LMI. Its Section 5 and Figure 1 explicitly describe approximate numerical solutions, allowing small affine and cone residuals. We address the exact feasibility question. The source's separate global theorem for matrix games uses a stepsize controlled by an instance-specific strictly complementary saddle point and a different proof; it is outside the present obstruction.

Conservative unconstrained AltGDA trajectories and their invariant quadratic energy are classical; see Bailey–Gidel–Piliouras [2, Theorem 4] and the symplectic analysis of Katona–Wang–Wibisono [3]. The interpolation ingredients in the source belong to the established performance-estimation framework [4, 5]. Our use of rotating planar modes concerns a different requirement: matching a finite collection of history Grams while enforcing genuine convex projections and a positive comparator gap.

2 The trajectory obstruction

Theorem 1. *For every integer $h \geq 0$ and every $\eta > 0$, there are compact convex polytopes X, Y in the unit ball of $\mathbb{R}^{2h+9}$, a symmetric positive definite matrix A with $\|A\| = 1$, a feasible initialization, and a feasible comparator such that the actual iteration* (1)*, with the initial padding above, satisfies*

$$(H_2^x)^T H_2^x = (H_1^x)^T H_1^x, \qquad (H_2^y)^T H_2^y = (H_1^y)^T H_1^y, \qquad g_1 > 0. \tag{6}$$

Consequently no pair of symmetric matrices in (4) *can make* (5) *hold on every actual window in this class. No assumption $V_k \geq 0$ is needed for this conclusion.*

The contradiction follows directly from (6): every such potential has $V_2 = V_1$. The content is the construction of an initialized projected trajectory with these simultaneous properties. We give it explicitly below. The constructed rotating segment is finite; its continuation on the same sets need not remain a rotating segment.

2.1 A finite moment system

Lemma 1. *Given an integer $K \geq 1$ and $\eta > 0$, one can choose $M = K + 1$ distinct angles $0 < \theta_r < \min\{\eta/4, \pi/2\}$ and real coefficients c_r such that*

$$m_j := \sum_{r=1}^{M} c_r \sin((j + \tfrac{1}{2})\theta_r), \qquad m_0 = 1, \quad m_j = 0 \quad (1 \leq j \leq K). \tag{7}$$

Proof. Put $b = \min\{\eta/4, \pi/2\}$ and $\theta_r = rb/(M + 1)$. The identity

$$\frac{\sin((j + 1/2)\theta)}{\sin(\theta/2)} = 1 + 2\sum_{q=1}^{j} \cos(q\theta) =: P_j(\cos\theta)$$

expresses P_j as a polynomial of degree j with leading coefficient 2^j; $P_0 = 1$. Thus the matrix with entries $\sin((j + 1/2)\theta_r)$, for $0 \leq j \leq K$ and $1 \leq r \leq M$, has determinant

$$2^{K(K+1)/2} \prod_{r=1}^{M} \sin(\theta_r/2) \prod_{r<s} (\cos\theta_s - \cos\theta_r) \neq 0.$$

Solving this finite real system with right side $(1, 0, \ldots, 0)^T$ gives the coefficients. No estimate of its conditioning is needed. ∎

2.2 Rotating states and exact normals

Fix h, η and set

$$H = h + 2, \qquad K = h + 3, \qquad M = h + 4.$$

Use Lemma 1 and, for $1 \leq r \leq M$, define

$$\begin{gathered} t_r = 2\sin(\theta_r/2), \qquad \epsilon_r = \begin{cases} -1, & c_r < 0, \\ 1, & c_r \geq 0, \end{cases} \qquad b_r = \frac{\epsilon_r t_r}{2}, \\ s_r = t_r + b_r, \qquad a_r = \frac{s_r}{\eta}, \qquad w_r = \frac{2|c_r|}{t_r}. \end{gathered} \tag{8}$$

Then $w_r \geq 0$, $w_r b_r = c_r$ and

$$0 < s_r \leq \tfrac{3}{2} t_r < \tfrac{3}{8}\eta, \qquad 0 < a_r < \tfrac{3}{8}. \tag{9}$$

The convention at $c_r = 0$ gives $w_r = 0$ and causes no exception.

Decompose $\mathbb{R}^{2h+9}$ into one scalar coordinate and M mutually orthogonal planes. Write $u(\phi) = (\cos\phi, \sin\phi)$ and let

$$A = \operatorname{diag}(1, a_1 I_2, \ldots, a_M I_2). \tag{10}$$

It is symmetric positive definite with norm one. For every integer i, define algebraic states

$$\begin{aligned}\xi_i &= \big(1, (\sqrt{w_r}\, u(i\theta_r))_{r=1}^{M}\big),\\ \zeta_i &= \big(0, (\sqrt{w_r}\, u(i\theta_r + \theta_r/2 - \pi/2))_{r=1}^{M}\big),\end{aligned} \tag{11}$$

and a comparator

$$\xi_\star = (1, 0, \dots, 0), \qquad \zeta_\star = (\gamma, 0, \dots, 0), \qquad \gamma = \frac{1}{2\eta}. \tag{12}$$

Use different symbols for these algebraic states to keep their indices distinct from the initialized trajectory below.

Define residuals by the actual update expressions

$$\alpha_i = \xi_{i-1} - \eta A\zeta_{i-1} - \xi_i, \qquad \beta_i = \zeta_{i-1} + \eta A\xi_i - \zeta_i.$$

Their scalar coordinates are 0 and η, respectively, and their planar coordinates are

$$(\alpha_i)_r = \sqrt{w_r} b_r u(i\theta_r + \pi/2 - \theta_r/2), \qquad (\beta_i)_r = \sqrt{w_r} b_r u(i\theta_r). \tag{13}$$

For example, identifying a plane with $\mathbb{C}$, the two update identities reduce to

$$e^{-\mathrm{i}\theta} - 1 + \mathrm{i}se^{-\mathrm{i}\theta/2} = \mathrm{i}(s-t)e^{-\mathrm{i}\theta/2}, \qquad -\mathrm{i}e^{-\mathrm{i}\theta/2} + s + \mathrm{i}e^{\mathrm{i}\theta/2} = s - t,$$

where $t = 2\sin(\theta/2)$ and $s - t = b$. This verifies both signs in (13).

For any integers i, j and $\ell = i - j$, direct inner products now give

$$\langle \alpha_j, \xi_i - \xi_j\rangle = \langle \beta_j, \zeta_i - \zeta_j\rangle = m_\ell - 1. \tag{14}$$

Here the formula defining m_ℓ in (7) is used also for negative indices. Since $m_{-r} = -m_{r-1}$ for $r \geq 1$, the prescribed moments imply, whenever $|\ell| \leq K$,

$$m_\ell - 1 = \begin{cases} 0, & \ell = 0,\\ -2, & \ell = -1,\\ -1, & 1 \leq \ell \leq K \text{ or } -K \leq \ell \leq -2.\end{cases} \tag{15}$$

The comparator inequalities are

$$\langle \alpha_j, \xi_\star - \xi_j\rangle = -1, \qquad \langle \beta_j, \zeta_\star - \zeta_j\rangle = \eta\gamma - 1 = -\tfrac{1}{2}. \tag{16}$$

These exact negative values are the reason for using signed moments in the construction.

2.3 One fixed pair of convex sets and a valid start

Define the fixed polytopes

$$C_X = \mathrm{conv}\{\xi_\star, \xi_{-1}, \xi_0, \dots, \xi_H\}, \qquad C_Y = \mathrm{conv}\{\zeta_\star, \zeta_{-1}, \zeta_0, \dots, \zeta_H\}. \tag{17}$$

For $0 \le j \le H$, every sampled vertex index in $\{-1, 0, \dots, H\}$ differs from j by at most $H + 1 = K$. Equations (14)–(16), extended from vertices by linearity, therefore prove

$$\alpha_j \in N_{C_X}(\xi_j), \qquad \beta_j \in N_{C_Y}(\zeta_j), \qquad 0 \le j \le H.$$

The projection characterization $z = P_C(v)$ if and only if $v - z \in N_C(z)$ and $z \in C$ gives

$$\xi_j = P_{C_X}(\xi_{j-1} - \eta A\zeta_{j-1}), \qquad \zeta_j = P_{C_Y}(\zeta_{j-1} + \eta A\xi_j), \qquad 0 \le j \le H. \tag{18}$$

Start the actual algorithm at $(x_0, y_0) = (\xi_{-1}, \zeta_{-1})$ and pad its initial residuals by zero. It follows that

$$(x_t, y_t, \bar{d}_t, d_t) = (\xi_{t-1}, \zeta_{t-1}, \alpha_{t-1}, \beta_{t-1}), \qquad 1 \le t \le H + 1. \tag{19}$$

Thus the generic window with algebraic indices $0, \dots, H$ is reached at actual base time one. Its nonzero base residuals are generated by the preceding projection, not substituted for the initial padding. The extra predecessor in (17) and the moment at index $K = H+1$ enforce this reachability.

Let

$$O = \operatorname{diag}(1, R(\theta_1), \dots, R(\theta_M)),$$

where $R(\theta)$ is planar rotation. This orthogonal matrix commutes with A, fixes both comparator vectors and their images, and maps each algebraic state, residual and operator image at index i to the corresponding vector at index $i + 1$. Consequently the exact histories of the actual trajectory satisfy

$$H_2^x = OH_1^x, \qquad H_2^y = OH_1^y.$$

All columns involved are within (19): the largest residual and p index is the algebraic $H = h + 2$. Both Gram equalities in (6) follow. At the same base time,

$$g_1 = \langle A\zeta_\star, \xi_0\rangle - \langle A\xi_\star, \zeta_0\rangle = \gamma > 0.$$

Finally put $W = \sum_r w_r$, $D = \max\{\sqrt{1 + W}, \gamma\}$ and $\rho = 1/D$. Scale every state, comparator and set above by ρ, keeping A and η unchanged. Euclidean projection commutes with this common positive scaling. The residuals scale by ρ, the Gram equalities persist, and the gap becomes $\rho^2\gamma > 0$. The two polytopes now lie in the unit ball. This proves Theorem 1 for every $h \ge 0$, including $h = 0$, and every $\eta > 0$.

3 Exact infeasibility of the proposed LMI

Corollary 3.1. *For every finite integer $h \ge 0$ and every $\eta > 0$, the exact Lyapunov LMI in Appendix B.2, equations (B.1)–(B.2), of arXiv:2609.32995v1 is infeasible. Its C1 and C3 subsystem is already infeasible; adding C2 cannot change this conclusion.*

Proof. The source introduces an auxiliary quadratic expression $\mathcal{R}_k$ on the full window and separately certifies

$$V_{k+1} - V_k + \mathcal{R}_k \le 0 \quad \text{(C1)}, \qquad g_k - \mathcal{R}_k \le 0 \quad \text{(C3)}.$$

Adding them gives (5) on every actual window with $\|A\| \le 1$. Theorem 1 excludes that inequality for the exact source histories.

For completeness, this implication can be checked directly from the signs of the source's matrix identities. For a base window, let

$$\begin{aligned} \mathsf{X} &= [x_\star \mid x_k \mid \cdots \mid x_{k+h+2}], \quad \mathsf{P} = A\mathsf{X}, \\ \mathsf{Y} &= [y_\star \mid y_k \mid \cdots \mid y_{k+h+1}], \quad \mathsf{Q} = A^T\mathsf{Y}. \end{aligned}$$

Let G_x, G_y be the Gram matrices of the corresponding full source states, which include one more residual and operator-image column than (3). They are positive semidefinite. Add the C1 and C3 copies of the two affine identities and take their trace inner products with G_x, G_y. Their auxiliary matrices cancel; their free operator-equality multipliers cancel because $\mathsf{X}^T\mathsf{Q} = \mathsf{P}^T\mathsf{Y}$. The resulting exact equality is

$$\begin{aligned} V_{k+1} - V_k + g_k = \mathcal{N}_k &- \operatorname{tr}\big(U(\mathsf{X}^T\mathsf{X} - \mathsf{P}^T\mathsf{P})\big) \\ &- \operatorname{tr}\big(B(\mathsf{Y}^T\mathsf{Y} - \mathsf{Q}^T\mathsf{Q})\big) - \operatorname{tr}(Z_x G_x) - \operatorname{tr}(Z_y G_y). \end{aligned} \tag{20}$$

Here $U, B, Z_x, Z_y \succeq 0$ are sums of the respective C1 and C3 blocks. The term $\mathcal{N}_k$ is a nonnegative linear combination of inequalities

$$\langle \bar{d}_j, x_i - x_j \rangle \le 0, \qquad \langle d_j, y_i - y_j \rangle \le 0,$$

after including the comparator with zero normal, so $\mathcal{N}_k \le 0$. The two operator Gram differences in (20) are positive semidefinite for every actual norm-bounded A. Every term on the right is therefore nonpositive. At the base-time-one window of Theorem 1, the left side is strictly positive. This is a direct contradiction to the exact matrix identities. ∎

The source's C2 condition certifies $V_k \ge 0$. Neither that condition nor positive semidefiniteness of Q_x, Q_y is used in our contradiction. Conversely, the source's two auxiliary matrices are unrestricted symmetric matrices, so its C1/C3 certificate can be combined without losing feasibility: a combined certificate can be split by setting the auxiliary pair equal to the comparator-gap pair, assigning the combined multipliers to C1 and all zero multipliers to C3. Thus the obstruction does not arise from strengthening the published C1/C3 requirement.

The starred points in Appendix B are arbitrary feasible comparators. Requiring a saddle comparator instead would change the certificate and remove the direct relation to the supremum defining the duality gap. Likewise, the initial zero residual is a padding convention at actual time zero, not a restriction on every base window. Our use of the actual base-time-one window respects both source conventions.

4 What the obstruction leaves open

The construction proves that these finite histories cannot support the stated quadratic one-window comparator descent, for any positive stepsize. It does not assert that an arbitrary quadratic state function, a different history representation, an instance-dependent analysis, or a nonquadratic potential is impossible. Its dimension and geometry depend on the chosen finite history, and the positive gap can be small after normalization. The finite rotating segment has no asserted continuation with positive average gap. Therefore it does not disprove a dimension-independent C/T averaged-gap estimate at $\eta = 1/8$.

Theorem 2 of [1] is a conditional implication, whose exact feasibility hypothesis is excluded here for the specified LMI. The paper explicitly reports approximate numerical feasibility, rather than exact feasible matrices. Our conclusion is compatible with that description. We make no claim that the semidefinite system is weakly infeasible, no quantitative assertion about its distance to feasibility, and no diagnosis of a solver's tolerances. The separate matrix-game convergence theorem, and the formal verification of that theorem, are not challenged.

References

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